Test plan and execution record

> **v16.0 status.** The results in this plan are the v14.0 record unless stated. The v16.0 re-run is `docs/03-test-record.md` section K: K2 covers eight path scenarios, all COMPLETED. **TC-WRK-04 to TC-WRK-07 should be read as NOT RUN for v16.0** — the worker failure paths were attempted and produced no instance, for the reason recorded as K5.4. Two of the nine error codes have behavioural evidence on v16.0; the other seven are modelled, deployed and unproven.

> **Citations to files that are not in this tree.** An audit found 39 paths cited across 19 documents that no longer exist — the fourteen-pool model, the v1/v2 generators and the drawing scripts that went with them, and the Sprint 0 Node scaffold. Nothing was lost: all of it is in git history, and every citation is listed with what replaced it in 18-cited-files-no-longer-present.md.

Test counterpart to the delivered portfolio. The test strategy and the plan are in §1–§6; the execution record is in §7–§9. Every actual result and status in §7 is taken from `docs/03-test-record.md` or from the evidence file named in the row. Nothing is asserted as a pass without an artefact behind it. Where evidence is absent, the row says so.

Read with: `docs/03-test-record.md` (the working execution record this plan formalises), `docs/07-product-backlog.md` (the items being tested), `docs/02-status.md` (what is known not to work), `workers/README.md` and `workers/DEPLOYMENT.md` (worker behaviour and configuration).

---

1. Test strategy

1.1 Levels

Effort is layered for a reason. A defect is caught at the cheapest level that can catch it, and the expensive live runs are kept for integration work, not syntax.

Level	ID prefix	What it proves	Tooling	When it runs
Static model lint and layout analysis	`TC-BPMN`	The BPMN parses, the DI is complete, the drawer's rules hold, and the structural audit reports the census the design intends	`tools/validate_model.js`, `tools/analyse_layout.py`, `tools/analyse_bpmn.py`, `tools/audit_convergences.py`, Camunda Modeler 5.51.0	After every model regeneration
Worker unit and contract	`TC-WRK`	Each worker handles success, invalid input and the BPMN error conditions in its contract	`mvn` build plus throwaway models run against the live engine	After every worker change
Form render and validation	`TC-FRM`	Every form imports into the renderer that Tasklist uses, and the field types used are supported	`@bpmn-io/form-js` in headless Chrome	After every form regeneration
Deployment and configuration	`TC-DEP`	The model and its forms deploy together, within the batch ceiling, and re-deploy cleanly	`tools/deploy.sh`	At every release
End-to-end process	`TC-E2E`	The collaboration runs across pools with real messages, user tasks and worker jobs, with no incidents	`tools/demo_scenario.py`, Operate, Tasklist	At every release and after every behavioural change
Non-functional	`TC-NFR`	Data protection, role separation, audit behaviour, availability handling and accessibility intent	Static inspection of model, forms and worker sources; no dedicated NFR harness exists	At release, by inspection

1.2 Test types

Functional (process behaviour, exception routing, gateway outcomes), integration (message correlation, worker-to-engine, form-to-task binding), regression (preservation audit, re-run of both paths after a drawing-only change), static (parser, lint, DI completeness, job-type and error-code coverage), configuration (deployment, subscription, environment overrides), data-protection (card-data exclusion, minimum test data), usability/accessibility (form structure and descriptions). Negative testing forces a service failure, feeds invalid input and publishes an uncorrelated message.

There is no automated performance, load or penetration testing in this portfolio. That is a scope decision, not an omission, and it is stated again in §10.

1.3 Risk-based rationale: what is tested hardest

Rank	Area	Why it is tested hardest	Depth
1	Message correlation between pools	A missing or wrong correlation key produces a live incident (`JOB_NO_RETRIES`), and the whole collaboration depends on the hand-off working. It was the first real defect found by running the model (`docs/03-test-record.md` §B1)	Every throw event statically; both live paths end to end
2	Form field types and form-to-task binding	One unsupported field type made 17 fields and every form containing them unrenderable, and 38 user tasks depend on the forms. This was the single most valuable defect found (`docs/03-test-record.md` §F4)	All 37 forms rendered in the real renderer; 37 forms bound across 38 user tasks
3	Exception branches with an owner	The assessment rewards handling, not detection. A branch that reaches a bare end event is a failure of the modelled business process	All 9 error codes and 18 boundary events checked statically; the referral branch driven live
4	Layout quality	The rubric names overlap and parallel-adjacency as faults and asks for a diagram a person can read. It is a measured, automatable gate	Every rebuild measured; first two rows must be zero before the build is accepted
5	Preservation across a regeneration	R3 changes the drawing and two modelling rules; a silent loss of a form, candidate group, job type or exception hand-off would cost marks already earned	67 parsed checks against the previous file, run before R3 was accepted
6	Payment and financial handling	Money is involved; a duplicate charge or a lost confirmation must not be silently retried	Worker contract tests for every payment outcome; live happy path
7	Deployment size and versioning	Camunda 8 stores the whole BPMN resource per process definition, and **v14's fifteen processes mean fifteen copies**; the 4 MB append batch is a hard ceiling that already forced a design decision, and v14 has taken that risk on to connect the outside participants	Measured at every deployment; recorded in §7; the v14 exposure and its three mitigations are in `docs/11-white-box-external-participants.md` §5

Everything else (formatting, label wording, cosmetic clipping) is tested for existence and recorded, but is not allowed to hold a release.

---

2. Scope

In scope. The BPMN collaboration and its diagram interchange; the 36 Camunda Forms; the 36 Java worker classes and their 37 job-type subscriptions; the build and deployment scripts in `tools/` ; the two scripted run paths; the Operate and Tasklist evidence; and the configuration files that make the above reproducible.

Out of scope. Any external service; any production system; production identity and RBAC; persistent storage; the web applications themselves (Operate and Tasklist are used as supplied, not tested); and any cluster other than the local c8run instance.

2.1 Test items

Item	Version / identity	Location
BPMN model	`UFCEP6-0-3_Hospital_Patient_Pathway_v16.bpmn` — one executable process in the hospital pool plus six black-box participants (seven participants in all), `versionTag` 16.0.0	`model/`
Camunda Forms	37 files on 38 user tasks, form-js `schemaVersion` 19, `executionPlatformVersion` 8.10.0	`model/forms/`
External workers	`hospital-external-workers-1.0.0.jar` , 36 task classes, 37 subscriptions	`workers/`
Generator and gates	`build_v3.py` , `spec_v3.py` , `bpmn_builder_v2.py` , `bpmn_builder_v3.py` , `layout_engine.py` , `forms_spec.py`	`tools/`
Deployment and demo	`deploy.sh` , `run_v16.py` (14 scenarios), `render_v16.py` , `render_sections_v16.py`	`tools/`
Verifiers	`analyse_layout.py` , `analyse_bpmn.py` , `audit_convergences.py` , `forms_inventory.py` , `audit_accessibility.py` , `validate_model.js`	`tools/`
Evidence	run logs and UI screenshots	`screenshots/` , `diagram/`

> **The v2.0 items this table used to list are gone.** `_v2.bpmn` , `build_v2.py` , `spec_v2.py` , `render_diagram.sh` , `export_sections.py` and `verify_preservation.py` are not in this tree, and the 15-participant / 9-process / 36-form figures were the v2.0 collaboration, not this one. §3 below and the case tables record what was run against the shipped model; `docs/17` gives the v16 census with the numbers behind it.

2.2 Features to be tested

Message hand-off and correlation; the nine hospital processes and their user tasks; gateway behaviour (34 exclusive, 3 inclusive, 2 parallel, 2 event-based); the nine exception conditions; compensation definitions; the clinic-letter timer ladder; form rendering, binding and validation; worker success, validation failure and BPMN-error paths; deployment and re-deployment; both end-to-end paths; and the data-protection rule on card data.

2.3 Features NOT to be tested

Real provider integration; performance and load; concurrent multi-instance contention; security penetration of the c8run webapps; the Tasklist rendering path (blocked by an environmental `401` , see `TC-FRM-07`); WCAG conformance with assistive technology (`TC-NFR-06`); and any browser other than Chrome (`TC-NFR-07`).

The three branches that were never driven (`TC-E2E-03` to `TC-E2E-05`) and the live compensation trigger (`TC-E2E-06`) are no longer gaps: all four were driven on v14 (`docs/03-test-record.md` §J11–J15). What remains outstanding is Operate, Tasklist and Modeler screenshots of v14, and the two pre-existing documentation defects DEF-14 and DEF-16.

---

3. Entry, exit, suspension and resumption criteria

Entry criteria (before any test run).

#	Criterion	Check
E1	The engine answers: `GET /v2/topology` returns 200	c8run 8.10.0-alpha5 running on `localhost:8080` / `localhost:26500`

#	Criterion	Check
E2	The generator exits 0 and the two layout gates are zero	`python3 tools/build_v3.py` ; `python3 tools/analyse_layout.py model/UFCEP6-0-3_Hospital_Patient_Pathway_v16.bpmn`
E3	The model parses clean in the Modeler's parser	`node tools/validate_model.js model/…bpmn`
E4	The deployment is accepted	`bash tools/deploy.sh` reports 1 process definition and 37 forms
E5	The workers are running and subscribed	Start-up log shows the job-type subscriptions on `localhost:26500`
E6	The test data is seeded as intended	`application.yaml` demo switches set for the branch under test

Exit criteria (before a release is signed off).

#	Criterion
X1	Layout gates: sequence-flow overlaps 0 pairs and message-flow overlaps 0 pairs (`docs/03-test-record.md` §A1–A2)
X2	Preservation gate: 67/67 parsed checks pass for any release after R1
X3	Static parser gate: 0 warnings, 0 unresolved references, 0 dangling flows, 0 user tasks without a form or candidate group, 0 service tasks without a job type
X4	Both scripted paths complete with `incidents: none`
X5	Non-functional design checks (`TC-NFR-01` to `TC-NFR-04`) hold, or their failure is recorded as an accepted residual risk
X6	Every test row has an actual result and an evidence reference, or is honestly marked `NOT RUN` / `NOT EVIDENCED`

Suspension criteria. Suspend the run if the engine dies or the deployment is rejected; if a change makes either scripted path hang or raise an incident the defect is not attributed to the test; or if an expected evidence file cannot be written.

Resumption criteria. Resume only after the generator accepts the change (E2), the model parses clean (E3) and the change has been re-deployed (E4). A run made against an undeployed model is discarded, not recorded.

---

4. Test environment and configurations

Component	Configuration used	Where it is recorded
Engine	Camunda c8run **8.10.0-alpha5**, gRPC `localhost:26500` , REST `http://localhost:8080` ; `GET /v2/topology` → 200	`docs/01-gap-audit.md` header; `docs/03-test-record.md` header
Modeler	Camunda Modeler **5.51.0**; model target platform `Camunda 8.10 (alpha)` , target **Local C8Run**	`screenshots/camunda-modeler-v2.1-status-bar.png` ; `docs/03-test-record.md` §H5a
Model platform metadata	`modeler:executionPlatform="Camunda Cloud"` , `modeler:executionPlatformVersion="8.10.0"` , exporter 5.51.0	parsed from the model file
Model version tag	`zeebe:versionTag value="2.0.0"` on all processes (9 at v7.0, 15 at v14)	parsed from the model file
Workers runtime	Plain Java (no Spring Boot): `maven.compiler.release` **21**, `zeebe-client-java` **8.9.0**, Jackson 2.21.2, slf4j 2.0.17, logback 1.5.32	`workers/pom.xml`

Component	Configuration used	Where it is recorded
Build toolchain	Maven 3.9.9; built under JDK 25 while targeting release 21	`docs/01-gap-audit.md` header; `workers/pom.xml` comment
Form renderer	`@bpmn-io/form-js`, the renderer Tasklist uses, driven in headless Chrome	`docs/03-test-record.md` §F4
Diagram renderer	`bpmn-js` in headless Chrome, the same engine the Modeler canvas uses	`tools/render_v16.py`, `tools/render_sections_v16.py`
Browser	Google Chrome (macOS, `darwin-aarch64` bundle)	`tools/render_v16.py`; screenshot set
Scripting	Python 3 (`deploy.sh`, `demo_scenario.py`, the analysers); Node (the lint and export)	`tools/`

Configuration under test. The demo switches in `workers/src/main/resources/application.yaml` are part of the test configuration: `demo.simulated-failure` (`always` / `once`), `demo.payment.status` (`APPROVED` / `DECLINED` / `DUPLICATE` / `NO_CONFIRMATION`), `demo.payment.confirmation-lost-mode` (`bpmn-error` / `retry`), `demo.scheduling.force-no-slots`, `demo.external.unavailable-resources`, `demo.pathway.scenario` (`SEVEN_DAYS` / `ONE_MONTH` / `THREE_MONTHS`), and `demo.clock.today` / `offset-days`. Every key is overridable by an environment variable; the mapping is documented in the file itself and in `workers/DEPLOYMENT.md` §4.

---

5. Test data strategy

Data	How it is produced	Reproducibility
Patient reference	`demo_scenario.py` generates a per-run reference, e.g. `PAT-DEMO-18020` (normal path) and `PAT-DEMO-18106` (exception path)	Unique per run, so instances do not collide on correlation keys
Form variables	Filled by `demo_scenario.py` over the REST API exactly as a person would fill them in Tasklist: the same form fields, submitted with the same completion call	Deterministic per run
Dates and day counts	`DemoClock` is freezable and shiftable (`demo.clock.today`, `demo.clock.offset-days`) so that day counts and escalation bands are reproducible	Deterministic when the clock is pinned
Escalation bands	The in-memory store seeds an overdue clinic letter according to `demo.pathway.scenario` (`SEVEN_DAYS` → worst case 9 days; `ONE_MONTH` → 45 days; `THREE_MONTHS` → 120 days, the default)	Deterministic per scenario
External-service results	Simulated: `correspondence.dispatch-letter`, `scheduling.find-appointment-slots`, `external-resources.check-availability`, `payment.process-transaction`, `payment.process-refund` are deterministic stubs driven by the demo switches	Deterministic, but simulated
Failure/invalid input	`SimulatedFailureRegistry` can force `always` or `once` failure per job type; a canonical BPMN error code can be pinned as `job-type:ERROR_CODE`	Deterministic
Personal data	None. Test identifiers are synthetic (`PAT-DEMO-…`); no real patient data is used or required	By design (see `TC-NFR-08`)
Idempotency state	In-memory `HospitalStore`, reset when the worker JVM restarts	A repeat across a restart is not the same test as a repeat inside one JVM (a limitation, see §10)

Simulated external services and their limitations. The five simulated job types never call a network service. The four suppliers were black-box pools in v7.0, and from v14 they are white-box pools of their own. They genuinely exercise the success and failure branches, and they produce plausible references, but they cannot confirm delivery, settlement or capacity, nor reproduce the timing and partial-failure behaviour of a real provider. Swapping one for a real API means replacing one worker class (`workers/README.md`, "Simplifications and known gaps"). No test result in §7 should be read as evidence of real integration.

---

6. Defect management

How defects are logged. Every defect is found by running something, not by reading. It is entered against the test case that found it, with the symptom as observed, the cause, the fix, and the severity. The working record is §B and §F4 of `docs/03-test-record.md`; the table in §9 below formalises it with IDs and severities. **Severity scale.**

Severity	Meaning	Release effect
S1 Blocker	The release cannot be accepted: the model will not deploy or run, a scored deliverable would be lost, or a financial rule is broken	Must be fixed before the release
S2 Major	A scored behaviour is wrong or unreliable but a workaround exists	Must be fixed before the release unless explicitly accepted
S3 Minor	A visible quality defect that does not change behaviour	Fixed if time allows; recorded otherwise
S4 Cosmetic	Appearance only	Recorded; normally deferred

Triage rule. An S1 or S2 defect found during a run suspends the run under §3 until it is fixed and the run is repeated. S3 and S4 defects are recorded and carried; they do not suspend a run.

---

7. Test cases and execution record

Status legend: **PASS** (expected met, evidence present); **FAIL** (expected not met); **PARTIAL** (met in part, or met with an accepted deviation); **NOT RUN** (never executed); **NOT EVIDENCED** (claimed or implied somewhere but no evidence file exists in the workspace).

7.1 Static model, layout and structure: `TC-BPMN`

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-BPMN-01	No sequence-flow overlaps or parallel adjacency	Model regenerated	`python3 tools/analyse_layout.py model/…bpmn`	0 pairs / 0 px	0 pairs / 0 px (v2.1; v1.0 was 142 pairs / 18,951 px)	PASS	PB-05	`docs/03-test-record.md` §A1
TC-BPMN-02	No message-flow overlaps	as above	as above	0 pairs / 0 px	0 pairs / 0 px (v1.0 was 448 pairs / 507,620 px)	PASS	PB-05, PB-26	§A2
TC-BPMN-03	No diagonal segments	as above	as above	0 px	0 px	PASS	PB-05	§A3
TC-BPMN-04	No line passes through an unrelated shape	as above	as above	0 pairs	**0 pairs** in R3 (v7.0). v2.0 and v2.1 had 4: sibling boundary events sat 34 px apart on 36 px circles, and a line leaving the left one crossed the right one	**PASS**	PB-05, PB-34	§A4; `docs/06-diagram-engineering.md` §3.2–3.3
TC-BPMN-05	Bends per sequence flow ≤ 4	as above	as above	≤ 4	mean 1.1, max 4	PASS	PB-05	§A5

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-BPMN-06	A routing channel separates pools	as above	as above	present	104 px	PASS	PB-05, PB-26	§A6
TC-BPMN-07	Lanes exist where a pool covers several desks	as above	as above	lanes present	14 lanes across 6 pools at v7.0; **20 lanes at v14**, one for each of the six outside participants	PASS (re-measured)	PB-04	§A7; `docs/11` §4
TC-BPMN-08	Repetitive administration is contained	as above	as above	collapsed sub-processes used	3 collapsed sub-processes	PASS	PB-06	§A8
TC-BPMN-09	Crossings minimised	as above	as above	as low as possible	**94** in R3 (v7.0), down from 197 in v2.1, 147 in v2.0 and 42 in v1.0. Two causes: the router measures the crossings a candidate corridor would create instead of picking the nearest one, and giving every path its own end event removed the long runs to shared circles	**PASS**: 52% reduction, with 0 overlaps and 0 clipped lines held	PB-27, PB-34	`docs/06-diagram-engineering.md` §3.1, §3.4, §3.5
TC-BPMN-10	Model opens in Camunda Modeler with no errors	Modeler 5.51.0 installed	Open the model; read the status bar	0 errors	**0 errors** in v2.1 (`screenshots/camunda-modeler-v2.1-status-bar.png`). In R3 the underlying linter was re-run on the shipped file under the project's own `bpmnlint:recommended` config: **0 errors / 32 warnings**, against 39 errors in v2.1. DEF-18 is closed	**PASS** on both the Modeler and the stricter config	PB-29, PB-34	`screenshots/static-checks-v7.0.txt` §4; `docs/03-test-record.md` §I2b
TC-BPMN-11	Model parses clean in the Modeler's own parser	Node with `bpmn-moddle` and `zeebe-bpmn-moddle`	`NODE_MODULES_DIR=… node tools/validate_model.js model/…bpmn`	0 warnings, 0 unresolved refs, 0 dangling flows, 0 user tasks without a form or candidate group, 0 service tasks without a job type	exactly that	PASS	PB-24, PB-25	§H5d
TC-BPMN-12	Regeneration loses nothing scored	The v2.0 baseline is vendored at `tools/baseline/…bpmn`	`python3 tools/verify_preservation.py`	all checks pass	**67 checks, 67 passed, 0 failed** in R3 (62 in the v2.1 round, §H1)	PASS in R3, **not reproducible from this tree** — `verify_preservation.py` is not present, so the count cannot be re-run. What does ship and does run is the structural audit: `analyse_bpmn.py`, `analyse_layout.py`, `audit_convergences.py`	PB-25	`screenshots/static-checks-v7.0.txt` §2

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-BPMN-13	Sub-process steps remain in the file and executable	as above	Count inner elements and check reachability	28 inner elements present	28/28	PASS	PB-06, PB-25	§H1; `docs/05-v2.1-changes.md` §3
TC-BPMN-14	No Camunda 7 attributes	as above	Scan for the 6 forbidden attribute names	none present	none present	PASS	PB-25	§H1
TC-BPMN-15	Structural census matches the intended design	Model present	Parse participants, processes, lanes, tasks, forms, flows	15 pools, 9 executable processes, 14 lanes, 39 user tasks, 47 service tasks, 36 forms, 15 drawn message flows, 311 sequence flows, 65 end events (v7.0)	exactly those counts at v7.0; **v14: 15 pools, 15 executable processes, 20 lanes, 39 user tasks, 51 service tasks, 36 forms, 22 drawn message flows, 329 sequence flows, 72 end events**	PASS at v7.0; v14 counts re-parsed and listed in `docs/11` §4; **v16: 7 participants (1 executable process + 6 black-box), 12 lanes (3 responsibility groups holding 9 team lanes), 38 user tasks, 48 service tasks, 29 boundary events, 29 exclusive gateways, 37 forms, 15 message flows, 193 sequence flows, 22 end events**	PB-02, PB-04, PB-09, PB-13	parsed from `model/…bpmn` ; `README.md` headline numbers
TC-BPMN-24	No end event merges more than one terminating path	Model present	Count incoming sequence flows per end event	every end event has exactly one incoming flow	**65/65 have exactly one incoming; none is unreachable**	PASS	PB-35	`docs/06-diagram-engineering.md` §3.5
TC-BPMN-25	The drawn message flows are only the ones that inform	Model present	Compare the drawn set with the retention rule	every exception hand-off drawn, four system-boundary hand-offs drawn, nothing else	**15 drawn = 11 exception + 4 boundary; 100%**	PASS	PB-35	`docs/06-diagram-engineering.md` §3.6
TC-BPMN-16	All nine exception conditions exist and are reachable	Model present	Enumerate `bpmn:error` and boundary events	9 error codes, 18 boundary events, all attached to a host activity	9 and 18, matching names in `docs/05-v2.1-changes.md` §3	PASS	PB-07, PB-08	§H1; parsed from the model
TC-BPMN-17	The generator is deterministic	Dependencies installed	Run `python3 tools/build_v3.py` twice and compare	byte-identical output	byte-identical	PASS	PB-25	`docs/05-v2.1-changes.md` §5
TC-BPMN-18	Build-time guards refuse a dangerous model	Generator present	Inspect and exercise the guards	Refuses unbound/undefined forms, two elements in one grid cell, and a model mixing boolean `true` with text `"true"` comparisons	Guards present in `build_v3.py` (`check_cells` , form binding, `check_date_repair_points`) and the builder (boolean/text); the last two were added after defects `DEF-02` and `DEF-09` , and the date-repair guard after the v16 date work	PASS	PB-05, PB-15, PB-25	`tools/build_v3.py` ; `docs/03-test-record.md` §B2, §B9

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-BPMN-19	Every user task has a form and a candidate group	Model present	Parse `zeebe:formDefinition` and `zeebe:assignmentDefinition` on each user task	38/38 with both	38/38, candidate groups matching the pool roles	PASS	PB-09, PB-10	§H1, §H5d
TC-BPMN-20	Thinning the drawn message flows changed no behaviour	v2.1 deployed	Re-run both paths and compare incidents	both paths clean	both paths clean, `incidents: none` (hand-off is the throw event's job, not the drawn line)	PASS	PB-26	§H3
TC-BPMN-21	Message start events reduced to ≤ 2 per pool	—	Count start events	≤ 2 per pool	**43 start events**: target missed, because a Camunda 8 message start subscribes to one message name, so collapsing them needs a protocol change	FAIL (deferred to PB-33)	PB-33	§H6a; `docs/05-v2.1-changes.md` §4
TC-BPMN-22	Four clipped lines fixed	—	Re-run TC-BPMN-04	0 pairs	**0 pairs** in R3 (v7.0): sibling boundary-event pitch widened to `EVT + 6`, boundary-host rows grown by `BND_DEPTH`, and a verified `_unclip` pass that can only remove a defect	**PASS**	PB-34	`docs/06-diagram-engineering.md` §3.2–3.3
TC-BPMN-23	Sub-process boxes drawn open	—	Set `V2Builder.EXPAND_SUBPROCESSES = True` and measure	open boxes with no new faults	tried and reverted: 21 lines through a shape against 4, and 3 new overlaps	NOT RUN (deliberately reverted)	PB-06	§H6d; `docs/05-v2.1-changes.md` §4

7.2 Worker contract and integration (`TC-WRK`)

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-WRK-01	Workers build	Maven 3.9.9, offline repository populated	`mvn -q -o clean package`	BUILD SUCCESS	success, 29 MB fat jar	PASS	PB-13	§E1
TC-WRK-02	One subscription per job type at start-up	Engine up	Start the jar; read the log	subscription per job type	**37 job-type subscriptions** on `localhost:26500`	PASS	PB-13	§E2, §H2b
TC-WRK-03	The three v2.0 job types resolve	as above	Check the log for the new types	all three registered	`treatment.release-series`, `treatment.release-cycle-booking`, `treatment.record-capacity-retry` all registered	PASS	PB-18, PB-19	§E3
TC-WRK-04	Jobs complete with no incidents during the paths	Both paths run	Run `demo_scenario.py` and `--exception`; search incidents	no incidents	no incidents in either run	PASS	PB-13	§E4

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-WRK-05	Referral worker success and both error branches	Throwaway model in `/tmp`	Activate and exercise the worker	success, `REFERRAL_PACK_INCOMPLETE`, `REFERRAL_PACK_UNREADABLE` branches behave	all three exercised against the live cluster	PASS	PB-07, PB-13	`workers/README.md` ("Verified against the running engine")
TC-WRK-06	A blank correlation key fails safely	Throwaway model	Publish a message with a blank key	job fails three times, then an incident naming the variable; no BPMN error	exactly that	PASS	PB-03, PB-13	`workers/README.md`
TC-WRK-07	All domain job types run in one process	Throwaway 34-task model	Run it	completes with no incidents, including idempotency repeats	completed with no incidents	PASS	PB-13, PB-15	`workers/README.md`
TC-WRK-08	Failure injection works in both modes	Demo switches set	Force `always` and `once`, and the canonical simulated codes	BPMN error on the boundary in `always`; engine retry then success in `once`	both modes verified, plus empty-recipient dispatch and a malformed-variable job failure	PASS	PB-14	`workers/README.md`; `application.yaml`
TC-WRK-09	Idempotency	Live store	Repeat a booking/charge operation	same reference returned; no duplicate created	repeats return the same appointment; duplicate charges detected	PASS	PB-15	`workers/README.md`; §E1–E4
TC-WRK-10	An over-large refund does not throw an unroutable error	Throwaway model	Force a refund larger than the amount paid	failed job (incident), not a BPMN error	failed job, as designed: the model attaches no boundary error event to that step	PASS	PB-07	`workers/README.md` ("Simplifications and known gaps")
TC-WRK-11	Payment confirmation-lost handling is configurable	Demo switches	Run with `bpmn-error` and with `retry`	`bpmn-error` throws `PAYMENT_CONFIRMATION_LOST`; `retry` fails the job	both behaviours verified	PASS	PB-07, PB-14	`workers/README.md`; `application.yaml`
TC-WRK-12	Workers verified against the *full* collaboration	—	Run the workers against the generated collaboration	full integration verified at worker-test level	Not done at worker-build time: verification used equivalent throwaway models because the collaboration was still being regenerated; integration was later demonstrated by the two scripted paths (TC-E2E-01/02)	PARTIAL	PB-13	`workers/README.md` ("Not verified")
TC-WRK-13	The job-type table documents every model job type	Sources and README present	Compare model job types with the README table	every type documented	**Model has 35 distinct service-task job types; the README table documents 32.** The three v2.0 types are missing from it, and its summary line still reads "33 job workers = 1 + 32" although 36 task classes exist	FAIL (documentation defect `DEF-14`)	PB-13	`workers/README.md`; `workers/src/.../tasks/` (36 classes); §E3

7.3 Forms: `TC-FRM`

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-FRM-01	Every form imports and renders in the Tasklist renderer	`@bpmn-io/form-js` available; headless Chrome	Render all 37 forms	no failures	**`TOTAL=37 FAILED=0`**	PASS	PB-23	`screenshots/v16/forms-rendered-37.png` (re-run 2026-10-17); §F4
TC-FRM-02	No unsupported field types	as above	Census the field types actually emitted	only supported types	census: `textfield 69, radio 35, textarea 47, select 25, number 17, datetime 19, checkbox 12, checklist 6`, with no `date`; the earlier `"type": "date"` defect is fixed	PASS	PB-23	§F4
TC-FRM-03	Long forms are grouped by heading for accessibility	Forms present	Count `group` components	at least the long forms grouped	**15 `group` components across 7 forms; all 7 forms with eight or more fields are grouped, and no form below eight fields is ungrouped** — the grouping half of PB-12 was implemented after the v7.0 gap audit recorded it as missing	PASS	PB-12	form JSON census; `docs/01-gap-audit.md` §C3
TC-FRM-04	Fields carry an accessible description	Forms present	Count descriptions against fields	every field described	71 `description` values across 230 components, so coverage is partial	PARTIAL	PB-12	form JSON census
TC-FRM-05	Validation rules are present	Forms present	Count `validate` blocks and inspect rules	required/length/range where needed	186 `validate` blocks across the 37 forms	PASS	PB-11	form JSON census; `docs/01-gap-audit.md` §C2
TC-FRM-06	Every user task is bound to a defined form and vice versa	Model and forms present	Cross-check bindings and definitions	every bound form defined and every defined form bound	38/38 user tasks bound; 37 distinct forms, one shared on purpose (`answer-administrative-enquiry`, used by both administrative-enquiry tasks); build check asserts both directions	PASS	PB-09	`tools/build_v3.py` (`user_tasks`, binding checks); §H1
TC-FRM-07	Forms render inside Tasklist	c8run webapps running	Open a claimed task; watch the Task tab	the form renders	**NOT EVIDENCED**: the Task tab stayed empty; Tasklist gets `401` from `/v2/authentication/me` (with and without basic credentials), so it cannot fetch the schema. The forms themselves are proven to render by TC-FRM-01	NOT EVIDENCED (environmental, deferred to PB-32)	PB-32	§F5; `docs/02-status.md` ("Not done")
TC-FRM-09	A date that breaks the shared rule table returns to the form instead of parking the instance	c8run running, model v20 deployed, workers built from the same sources	Start the pathway with a future `referralDate`; complete the pack check; complete it again with a valid date	the first submit raises `INVALID_DATE`, no incident is left, the same user task reopens; the second submit continues	first submit: 0 incidents, a new `SEC_Task_CheckPack` in state CREATED, instance ACTIVE; second submit with 2026-01-05: no reopened task, 0 incidents, pathway continues	PASS	PB-09	`docs/21-date-rules-and-the-form-layer.md` §3.1; `tools/spec_v3.py check_date_repair_points`
TC-FRM-08	Labels use role-appropriate language	Forms present	Read the labels against the task's role	clerical tasks use administrative wording, clinical tasks clinical wording	not systematically checked; the gap audit records the concern (`docs/01-gap-audit.md` §C5) and no re-check exists	NOT EVIDENCED	PB-12	`docs/01-gap-audit.md` §C5

7.4 Deployment and configuration (`TC-DEP`)

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-DEP-01	Model and all forms deploy in one batch	Engine up	`bash tools/deploy.sh`	accepted, no error	**1 process definition + 37 forms**	PASS	PB-20	§C1
TC-DEP-02	Re-deployment after a change is accepted and versioned	as above	Re-run the deployment	accepted	accepted; the v2.1 rebuild deployed as **process definitions at v17 and forms at v18** (engine-assigned versions)	PASS	PB-20, PB-29	§C2, §H2a
TC-DEP-03	The resource stays inside the 4 MB append batch	as above	Measure the resource against the ceiling	inside the limit	v7.0: 397 KB × 9 processes, inside the limit. **v14 with all fifteen executable is rejected: "Can't append entry ... with size: 405002 this would exceed the maximum batch size ... currentBatchSize: 4095764". Nine fit, the tenth does not.** Shipped configuration: 9 executable in the main file (deploys clean, key 2251799813952610) + the 6 outside participants from a 30 KB second file (deploys clean)	**PASS as shipped; the 15-in-one-batch variant is a documented failure**	PB-20	§C3; `docs/04-modelling-decisions.md` §7.6; `docs/11` §5; `screenshots/static-checks-v14.txt` §8
TC-DEP-04	Workers build and subscribe on the stated runtime	JDK and Maven installed	Build and start	subscriptions on `localhost:26500`	37 subscriptions; the jar targets Java 21 bytecode and was built under JDK 25	PASS	PB-13	§H2b; `workers/pom.xml`

7.5 End-to-end paths and UI evidence: `TC-E2E`

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-E2E-01	Normal path across pools	Deployment and workers in place	`python3 tools/demo_scenario.py`	every step reached, no incidents	**all eleven steps reached**, `incidents: none` at each stage end; `medical-secretaries` , `consultants` , `outpatient-bookings` , `call-handling` , `treatment-bookings` and `finance-team` exchanged messages	PASS	PB-21	`screenshots/run-happy-path.log` ; §D1, §H3a
TC-E2E-02	Unreadable-referral exception path	as above	`python3 tools/demo_scenario.py --exception`	`REFERRAL_PACK_UNREADABLE` thrown, caught by the boundary event, case leaves the main line and joins the missing-information loop, no incident	exactly that, `incidents: none` ; the branch reaches `SEC_EGW_InfoRequest`	PASS	PB-22	`screenshots/run-exception-path.log` ; §D2, §H3b
TC-E2E-03	Declined-payment branch	Demo payment status set to `DECLINED`	Drive the path to payment	branch runs, no duplicate booking or charge	never executed	NOT RUN (deferred to PB-30)	PB-30	§D3, §G, §H6e; `docs/02-status.md`
TC-E2E-04	Payment-confirmation-lost branch	`DEMO_PAYMENT_STATUS=NO_CONFIRMATION` , `--investigate`	Drive the path to payment	investigation raised and a clinician asked to authorise urgent treatment	**executed on v14**: `FIN_Task_InvestigatePayment` and `CON_Task_AuthoriseUrgentTreatment` both driven, `incidents: none`	**PASS** (was NOT RUN)	PB-30	`screenshots/run-confirmation-lost-v14.log` ; `docs/03` §J12

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-E2E-05	Three-month letter escalation	`DEMO_LETTER_OVERDUE_DAYS=120` , `--overdue` , short-timer variant deployed	Drive the overdue-letter path	escalation to the Administrative Management Team	**executed on v14**: `SUB_Pathway_Report_Review` and `ADM_Task_ReferHigher` driven; with `DEMO_LETTER_OVERDUE_DAYS=45` the manager rung (`ADM_Task_ContactConsultant`) is driven instead	**PASS** (was NOT RUN)	PB-30	`screenshots/run-overdue-escalation-v14.log` , `run-escalation-manager-v14.log` ; `docs/03` §J13–J14
TC-E2E-06	Compensation triggered live	Engine and workers running, capacity forced unavailable, short-timer variant deployed	Force the giving-up path and observe the handler	compensation handler completes in a real instance	**executed on v14**: `TRT_Throw_CompSeries` , `TRT_Bnd_CompSeries` and `TRT_Comp_ReleaseSeries` all COMPLETED on instance 2251799813979308, no incident	**PASS** (was NOT RUN)	PB-31	`screenshots/run-compensation-v14.log` ; `docs/03` §J15
TC-E2E-07	A v2.0 task can be claimed and completed in Tasklist	c8run webapps	Open Tasklist, claim a task, complete it	task opens with its candidate group and form	**NOT EVIDENCED**: `docs/03-test-record.md` §F2 cites `screenshots/tasklist-camunda-form.png` for this, but no such file exists in `screenshots/` . What does exist is `screenshots/tasklist-open-tasks.png` , which shows the open task list and the pool names	NOT EVIDENCED	PB-32	`screenshots/tasklist-open-tasks.png` ; §F1, §F2, §H4d
TC-E2E-08	The engine reports no incidents and a completed instance shows its real path	Operate reachable	Read the dashboard and a completed instance	no incidents; executed steps and variables visible	**21 running instances, 0 with an incident**, "Your processes are healthy"; a completed Medical Secretaries instance shows its Instance History and the real variables `adminChecksCarriedOut` , `approvingClinician` , `authorizingClinician` , `chargeAmount` , `appointmentDate`	PASS	PB-21, PB-24	`screenshots/operate-dashboard.png` , `operate-instance-completed.png` , `operate-processes.png` ; §H4

> **Correction, v16.0.** The v14.0 result below cites `run-compensation-v14.log` , which contains no compensation at all. See the correction note in `docs/03-test-record.md` . v16.0 ships compensation and ran it: `screenshots/run-v16-compensation.log` .

7.6 Non-functional (`TC-NFR`)

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-NFR-01	No complete card data is stored or emitted	Payment worker sources	Inspect the variables the payment workers read and write	only a payment reference, status, date and amount	payment workers "never emit card data"; only `paymentReference` , `paymentStatus` , `paymentRef` , `paymentDate` , `paidAmount` are returned	PASS (design inspection)	PB-16	`workers/README.md` job types 13 and 24
TC-NFR-02	Clinical and financial responsibilities stay separate	Model and candidate groups	Check that clinical decisions sit in clinical groups and financial decisions in Finance, and that no user task mixes them	separation holds	gateway conditions route funding to `finance` , clinical decisions to `consultants` / `clinical-nurse-specialists` ; 38/38 candidate groups match the pool	PASS (static); behavioural proof of refusal NOT EVIDENCED	PB-10, PB-17	model `assignmentDefinition` ; `docs/04-modelling-decisions.md` §3; §H5d
TC-NFR-03	Audit records are not overwritten by clinical edits	Audit worker sources	Inspect the two audit workers	audit ref and timestamp written; clinical variables not written back; blank `patientRef` fails the job	exactly that	PASS (worker contract; engine-level immutability NOT TESTED)	PB-17	`workers/README.md` job types 2 and 3

ID	Objective	Preconditions	Steps	Expected	Actual	Status	PB	Evidence
TC-NFR-04	Behaviour during an external-service failure is defined	Demo switches	Force each simulated service to fail	the modelled boundary error fires with a defined owner; retries are bounded by `retries="3"`; no unbounded retry loop	each simulated code is reachable and its boundary event exists; retry cap 3 verified at worker level on throwaway models	PASS at worker level; full-path NOT RUN for the three undriven branches	PB-07, PB-14, PB-19	`workers/README.md`; §H1; §D3
TC-NFR-05	Performance and load	—	Measure throughput, latency or concurrency	no target exists	never measured; no NFR harness exists. One instance at a time is driven	NOT RUN	—	—
TC-NFR-06	Accessibility conformance	—	Run a WCAG conformance assessment with assistive technology	conformance level met	never run; only the structural checks TC-FRM-03/04 (both incomplete) were attempted	NOT RUN	PB-12	—
TC-NFR-07	Cross-browser and cross-platform behaviour	—	Repeat the form and UI checks in another browser	renders in more than Chrome	only Chrome was used	NOT RUN	—	§F4 (headless Chrome)
TC-NFR-08	No personal data in test artefacts	Repo present	Inspect test data and screenshots	synthetic identifiers only	`PAT-DEMO-…` references, illustrative financial figures, demo clock; no real patient data. A data-protection impact assessment was not produced	PASS for test data; DPIA NOT EVIDENCED	PB-16	`docs/04-modelling-decisions.md` §6 items 4–5; run logs

---

8. Requirement → test traceability matrix

Backlog item → the tests that exercise it. An item with no test beside it has not been tested.

The keys are the `Mj-nn` identifiers of the delivered workbook (`../../docs/GroupMj Product-Backlog.xlsx`). That workbook is the plan of record, so traceability runs against it and nothing else. Earlier revisions of this matrix used `PB-nn` keys from `07-product-backlog.md`, which the delivered workbook does not contain; `07-product-backlog.md` §11 is the crosswalk between the two schemes and maps each old key onto the `Mj-nn` row below.

Backlog item	Tests	Coverage note
Mj50 Test plan and gap audit	—	Analysis item; its own evidence is the measurement, not a test
Mj15 Participants, lanes and the executable model	TC-BPMN-15, TC-DEP-01	Counted and deployed
Mj28 Correlated hand-offs and deployment configuration	TC-WRK-06, TC-E2E-01, TC-E2E-02	Static + live
Mj15 Lanes inside multi-desk pools	TC-BPMN-07, TC-BPMN-15	Counted
Mj15 Layout engine	TC-BPMN-01, -02, -03, -05, -06, -18	Measured
Mj15 Collapsed sub-processes	TC-BPMN-08, -13, -23	Counted; open-box variant deferred
Mj16, Mj21, Mj23 Exception branches with owners	TC-BPMN-16, TC-WRK-05, TC-WRK-10, TC-E2E-02, TC-NFR-04	One branch live; others static
Mj25 Letter timers and escalation	TC-BPMN-16, TC-E2E-05	**Driven on v14**, both rungs (short-timer variant; `docs/03` §J13–J14)

Backlog item	Tests	Coverage note
Mj36–Mj44 Forms per user task	TC-BPMN-19, TC-FRM-01, TC-FRM-06	Complete
Mj15, Mj46 Candidate groups	TC-BPMN-10, TC-BPMN-19, TC-NFR-02	Static
Mj45 Form validation	TC-FRM-05	Structural
Mj45 Accessibility and descriptions	TC-FRM-03, TC-FRM-04, TC-FRM-08, TC-NFR-06	**Superseded on v16.0**: 237/237 fields described and 15 group components; see `docs/20` and `screenshots/v16/accessibility-report.json`
Mj29–Mj34 Workers per job type	TC-WRK-01 – TC-WRK-04, TC-WRK-07, TC-WRK-12, TC-WRK-13, TC-DEP-04	Integration level partial
Mj29–Mj33, Mj53 Simulated services and failure switches	TC-WRK-08, TC-WRK-11, TC-NFR-04	Complete at worker level
Mj22, Mj33 Duplicate suppression	TC-WRK-07, TC-WRK-09, TC-E2E-01	Verified
Mj41, Mj48 No card data	TC-NFR-01, TC-NFR-08	Inspection
Mj47 Audit workers	TC-NFR-02, TC-NFR-03	Inspection
Mj20 Compensation	TC-BPMN-16, TC-E2E-06	Modelled and **live trigger driven on v16.0** (`screenshots/run-v16-compensation.log`)
Mj20, Mj33 Capped external-capacity retry	TC-WRK-08, TC-NFR-04	Worker level
Mj28, Mj58 Deployment	TC-DEP-01, -02, -03	Complete
Mj52, Mj56 Normal path evidence	TC-E2E-01, TC-E2E-08	Complete
Mj52 Exception path evidence	TC-E2E-02	Complete
Mj44, Mj53 Form rendering	TC-FRM-01, TC-FRM-02	Complete
Mj50, Mj51, Mj55 Test record and decisions	TC-BPMN-11, TC-E2E-08	The record is evidence, not a subject
Mj55 Preservation audit	TC-BPMN-12, -13, -14, -17, -18	Complete
Mj15 Message-flow thinning	TC-BPMN-02, -20, TC-E2E-01, TC-E2E-02	Complete
Mj15 End-event merge	TC-BPMN-09	Trade-off measured
Mj15 Lane consolidation	TC-BPMN-07, TC-BPMN-15	Counted
Mj54, Mj57 Modeler validation and export	TC-BPMN-10, -11	Complete
Mj52 Undriven branches	TC-E2E-03, -04, -05	**NOT RUN**
Mj20, Mj52 Live compensation	TC-E2E-06	**Driven on v16.0**: handler completed in a real instance, no incident
Mj45, Mj54 Tasklist form display	TC-FRM-07, TC-E2E-07	v16.0 evidence: `screenshots/v16/tasklist-form-rendered.png` ; `docs/20`
Mj15 Start-event reduction	TC-BPMN-21	**FAIL (deferred)**
Mj15, Mj57 Crossing reduction	TC-BPMN-04, -09, -22	**PASS in R3**: 197 → 94 crossings, clipped lines 4 → 0
Mj15 Readable terminals and hand-offs	TC-BPMN-24, -25	**PASS in R3**: one end event per terminating path; 15 drawn message flows, with no exception hand-off lost
— Real provider integration	—	Explicitly out of scope. The workbook has no row for it; the deferred item is `PB-52` in `docs/07` §11.1

9. Defect log

9.1 Defects recorded in `docs/03-test-record.md`

All thirteen were found by running the model or the forms, not by reading it.

ID	Test	Severity	Symptom as observed	Cause	Fix / status
DEF-01	TC-WRK-06	S1	`JOB_NO_RETRIES` incident on the first message throw	A worker publishes the message and needs the correlation key as a process variable	Builder now writes a `correlationKey` input mapping onto every throw event. **Fixed**
DEF-02	TC-E2E-01	S1	Incident at `CON_GW_ConsentGiven` : *Can't compare* `"true"` *with* `true`	Yes/No radios submit the text `"true"` ; ten conditions compared against a boolean, giving a FEEL null	All ten conditions compare against `"true"` ; the build refuses a model that mixes the two. **Fixed**
DEF-03	TC-WRK-04	S1	`booking.record-appointment-outcome` failed	It ran before the form that captures the outcome	Task order swapped; **Fixed**
DEF-04	TC-E2E-01	S2	`SUB_CallHandling_Contact` forked: two tasks active at once	A node inside the collapsed sub-process had two unconditional outgoing flows	Exclusive gateway added inside the sub-process with a condition and a default. **Fixed**
DEF-05	TC-BPMN-03	S4	Lines drawn as long diagonals	Same-row links used port offsets, tilting a 300 px run by 7 px	Straight runs take a single level. **Fixed**
DEF-06	TC-BPMN-04	S3	Every flow out of a boundary event routed to the far-left channel	Boundary events inherited `col = 0` ; the host's grid cell was applied after routing was planned	Boundary events inherit their host's cell during the row phase. **Fixed**
DEF-07	TC-BPMN-02	S3	Message risers landing on top of each other	All pools share a column grid, so channel 5 in one pool sat at the same x as channel 5 in another	Message flows get their own half of every channel, allocated globally per column. **Fixed**
DEF-08	TC-BPMN-02	S3	Riser lookup collapsing	One message has two risers that can share a column, so keying by message index merged them	The lookup key identifies the segment, not the message. **Fixed**
DEF-09	TC-BPMN-18	S2	Two new elements drawn on top of existing ones	Compensation handlers and the retry chain were given grid cells already in use	Cells remapped; `build_v2.py` refuses a model where two elements share a cell. **Fixed**
DEF-10	TC-BPMN-04	S4	Message lines running behind the note boxes	Notes sat at the foot of each pool, in the path of the risers	Notes moved into one panel below the collaboration. **Fixed**
DEF-11	TC-E2E-06	S1	Compensating for a booking that was never made	Compensation handlers were attached before the confirm step in one place	Handlers attached to the activities that actually commit the booking. **Fixed**
DEF-12	TC-FRM-01	S1	`render error: form field of type <date> not supported` ; the Tasklist Task pane stayed blank	17 fields across the forms used `"type": "date"` ; Camunda Forms has no `date` type, so a date-only field is `datetime` with `subtype: "date"`	`tools/forms_spec.py` now emits `datetime` with a subtype; all 36 forms render. **Fixed**
DEF-13	TC-FRM-01	S1	The DEF-12 fix silently did not reach the generated files	`build_v2.py` imported `spec_v2` before `forms_spec` , and `spec_v2` put the v1.0 tools directory on `sys.path` , so the v1.0 `forms_spec.py` was loaded	Import order corrected; **Fixed**

9.2 Documentation and planning defects found while writing this plan

These are not model defects. They are recorded because they affect a marker's reading of the portfolio and are the kind of thing configuration management is supposed to catch.

ID	Severity	Finding	Evidence	Status
DEF-14	S3	`workers/README.md` documented 32 of the 35 domain job types the model uses and its summary line was stale	compare `workers/README.md` table with the model job types and `workers/src/.../tasks/` (36 classes)	**Closed in R3.** The catalogue now lists 36 rows (0–35), names the three missing types (`treatment.record-capacity-retry`, `treatment.release-series`, `treatment.release-cycle-booking`), and its summary lines read 36 workers = 1 dispatcher + 35 service tasks. See FB-09.
DEF-15	S2	`docs/06-visual-compaction.md` described a compaction (9 lanes, 12 sub-processes, 15 message flows, canvas 2,398 × 6,858) that was not present in the shipped model	parse the model file; compare with `README.md`, `docs/02-status.md`, `docs/03-test-record.md` §H, `docs/05-v2.1-changes.md` §2	**Closed in R3**: the drifted document was deleted and replaced by `docs/06-diagram-engineering.md`, which describes the shipped drawing and records the abandoned pass with its measurements. See FB-11.
DEF-16	S3	The subscription count was stated as 34 in `workers/README.md` and 37 in `docs/03-test-record.md` §E2	36 task classes; 35 distinct model job types; one alias; `publish-message` → 37 subscriptions	**Closed in R3.** `workers/README.md` now states 37 subscriptions and shows the arithmetic. This plan documents the same figure in `docs/07-product-backlog.md` A5.
DEF-17	S3	`docs/03-test-record.md` §F2 cited `screenshots/tasklist-camunda-form.png`, which did not exist	directory listing of `screenshots/`	**Closed in R3.** The Tasklist login answered on a clean engine, a user task was left pending, and the form was captured open in Tasklist (`screenshots/tasklist-camunda-form.png`, landed on `/tasklist/2251799813689594`). See `TC-E2E-07`
DEF-18	S4	Under the project's own `bpmnlint:recommended` config, two elements were implicit starts: `TRT_Throw_BookingPending` (the head of the counted capacity retry, left unreachable when the v2.0 pass pointed the capacity boundary event straight at the release step) and `SUB_Secretaries_Dispatch_Chase` (the retry step inside the collapsed dispatch sub-process)	`bpmnlint model/*.bpmn`; walk the sequence flows into each element	**Closed in R3**: the capacity boundary now enters the retry chain it was built for, and the dispatch sub-process gained an inner gateway that reaches the retry step on the status the worker already returns. `bpmnlint` reports 0 errors; both paths re-run with `incidents: none`
DEF-19	S3	Outpatient Bookings parked at `OUT_Catch_PhoneContact` for ever and Call Handling always had a contact task pending	polled for 90 s; re-ran with the message published by hand	**Closed in R3: two causes, neither of them the model's message wiring.** (1) **Test harness:** `wait_for_task` filtered user tasks by process definition *id*, so on a shared engine it regularly completed a matching task belonging to an abandoned instance from an earlier session. The run's own instance then waited for ever, and the branch looked broken. The lookup is now scoped to the instances this run creates, and the branch completes. (2) **Model:** the Call Handling process ended by looping straight back into its own sub-process (`SUB_CallHandling_Contact → CALL_Throw_ContactFinished → SUB_CallHandling_Contact`), so it never finished and always had a pending contact task. It now ends after the hand-off is published. Proven correct in the strongest available way: publishing `appointment.phone-contact-finished` by hand with the right key moved a stuck instance immediately, so correlation was never the problem
DEF-22	S3	The capacity retry could never finish: `TRT_Auto_RecordRetry` mapped `=externalResourceAttempts + 1`, and the counter starts unset, so `null + 1` is null. `TRT_GW_CapacityCap` then evaluated `externalResourceAttempts < 3` to NULL rather than BOOLEAN and raised an incident, so the loop neither retried nor released and the compensation handlers were unreachable	ran the short-timer variant with `external-resources.check-availability` forced to fail; two treatment instances parked at `TRT_GW_CapacityCap` with incident "*Expected result of the expression 'externalResourceAttempts < 3' to be 'BOOLEAN', but was 'NULL'*"	**Closed in R3**: the mapping is now `=if externalResourceAttempts = null then 1 else externalResourceAttempts + 1`. Re-run: the retry loop counts 1, 2, 3, takes the cap, and **the compensation handlers execute** (`TRT_Comp_ReleaseSeries` completed 4 times). That is the first live execution of compensation this portfolio has had
DEF-23	S4	After the higher-management rung was worked, `ADM_Throw_HigherRecorded` parked: the "record the escalation" throw did not complete	active element instances for administrative-management	**Closed in R3.** Same family as DEF-21: Administrative Management is *started by* the escalation, so it inherits the batch context and has no `patientRef`, but its two "record the escalation" throws still keyed on the patient. Both now use `="letter-escalation"`. Confirmed on a clean engine: the ladder runs to both rungs and the Administrative Management instance reaches `COMPLETED` (`ADM_Throw_HigherRecorded` published `letter.escalation-recorded`, no active elements left)

ID	Severity	Finding	Evidence	Status
DEF-21	S3	The clinic-letter escalation ladder cannot complete. `PCW_Start_WeeklyReview` is a **timer** start event, so the instance has no `patientRef`, yet `PCW_Throw_EscalateManager` and `PCW_Throw_EscalateHigher` key their messages on `=patientRef + "-letter-escalation"`. The key evaluates to null, so the worker refuses the job: "*Process variable 'correlationKey' is missing or blank for message 'letter.escalation-higher-management'*". The instance parks. Found by demonstrating the branch, which is what the demonstration was for	ran the short-timer variant (`model/demo/`), drove the weekly review task (`SUB_Pathway_Report_Review` completed), then read the worker log: three failed `publish-message` jobs on `PCW_Throw_EscalateHigher`, remainingRetries 2/1/0	**Closed in R3.** Both escalation messages and their `bpmn:message` subscriptions now correlate on `="letter-escalation"`, the review run rather than a patient the timer-started instance does not have. Re-run: `PUBLISHED ... letter.escalation-higher-management correlationKey=letter-escalation`, and `ADM_Task_ReferHigher` was driven to completion. The compromise is that all escalations from a review run share one key, so only one Administrative Management instance exists at a time; a production design would key on the letter, and that is written down rather than hidden
DEF-20	S4	The clinic-letter escalation ladder and the compensation handlers cannot be driven in a run of demonstration length: the weekly review is a `R/PT168H` timer start event and the capacity retry waits on `P3D` timers	traced the pathway process: `PCW_Start_WeeklyReview` is a timer start event, not a message start; `TRT_Catch_RetryTimer` is `P3D`	**Closed in R3.** A short-timer variant (`model/demo/`, three timers set to `PT15S`, identical process ids) is built by `build_v2.py` and deployed with `BPMN=model/demo/... ./tools/deploy.sh`. The shipped model keeps the real durations. Both were then driven live: the ladder runs to both rungs and completes, and the compensation handler executes. Note the side effect recorded in `model/demo/README.md`: the variant's repeating timer cannot be withdrawn through the 8.10.0-alpha5 API and needs an engine restart

10. What is not covered, and the residual risk

Not covered	Why	Residual risk	Where it is tracked
Declined payment, confirmation lost, three-month escalation	Not driven before the release was signed off	A branch that is modelled but never run can still contain a routing or data defect; the payment branches carry financial consequence	PB-30; `docs/03-test-record.md` §D3, §H6e; risk R-05
Live compensation	No run gave up a committed resource	The model-side compensation wiring is asserted statically but its runtime behaviour is unproven	PB-31; risk R-06
Tasklist rendering	c8run authentication returns `401` for this setup	The forms are proven to render, but the delivered user journey through Tasklist is not demonstrated	PB-32; `docs/03-test-record.md` §F5; risk R-07
Real external services	Five simulated job types	Delivery, settlement and capacity behaviour of real providers is unknown, including partial failure	risk R-04; `workers/README.md`
Performance, load and concurrent instances	No NFR harness and no target was set	Single-instance timing proves nothing about the audit trail under concurrent pathway load	TC-NFR-05; risk R-08
Accessibility conformance	Only structural field checks were attempted, and the grouping half of PB-12 was not delivered	The forms may not meet the accessibility expectation in the rubric despite rendering correctly	TC-FRM-03, TC-FRM-06, TC-NFR-06; risk R-09
Security of the webapps and the cluster	Out of scope; local c8run only	Not a deliverable, but any real deployment would need it	`docs/09-risk-contingency-and-config-management.md` §1
Cross-instance compensation	Camunda 8 compensation does not cross process instances; a refund is an explicit hand-off instead	The design limitation is deliberate and documented; it is not tested as a compensation	`docs/04-modelling-decisions.md` §5
Message start events ≤ 2 per pool	Deferred with the arithmetic recorded	The start-event target in the brief remains unmet; it needs a message-protocol change	TC-BPMN-21; PB-33; risk R-10
Crossing reduction and clipped lines	**Delivered in R3**	197 → 94 crossings, 4 → 0 clipped lines, 0 overlaps held	TC-BPMN-04, -09, -22; PB-34
One end event per path and far fewer dashed lines	**Delivered in R3**	32 → 66 end events, none merged; 31 → 15 drawn message flows with nothing meaningful dropped. **v14 trades four of those lines back (15 → 22) for six connected outside participants**, the deliberate exception, recorded in `docs/11`	TC-BPMN-24, -25; PB-35

Not covered	Why	Residual risk	Where it is tracked
Idempotency across a worker restart	State is in memory in one JVM	A restart could allow a repeat that a persistent store would prevent	`docs/04-modelling-decisions.md` §7 item 5; risk R-11

---

11. Assumptions

1. **A1: Worker runtime.** The brief supplied to the author of this document describes the workers as "Java 17 Spring Boot". The artefact does not match: `workers/pom.xml` sets `maven.compiler.release` to **21**, depends on `io.camunda:zeebe-client-java` 8.9.0, and contains no Spring Boot dependency; configuration is plain YAML read by `AppConfig`. This plan tests the worker as it is built.
2. **A2: Test dates and runner.** No test-run dates, machine names or operator identities are recorded beyond the environment header in `docs/03-test-record.md`. Test runs are attributed to first/second owners in `docs/07-product-backlog.md`, which is a planning value, not a recorder's log.
3. **A3: "37 job types".** The model defines 47 service tasks across 35 distinct job types; the workers produce 37 job-type subscriptions (35 model types + one alias + `publish-message`). Where a document says "37 job types" or "34 subscriptions", the reconciliation is in `docs/07-product-backlog.md` Assumption A5 and defect DEF-16.
4. **A4: Diagram document.** This plan takes the shipped model file as the source of truth (14 lanes, 3 sub-processes, 15 drawn message flows, 65 end events, canvas 5,332 × 14,176). The earlier compaction document that disagreed with it was removed in R3; DEF-15 is closed.
5. **A5: Missing screenshot.** `screenshots/tasklist-camunda-form.png`, cited by `docs/03-test-record.md` §F2, does not exist in the workspace. `TC-E2E-07` is therefore recorded as `NOT EVIDENCED` rather than as a pass.
6. **A6: Accessibility.** No accessibility rationale document exists in `docs/`, although `docs/01-gap-audit.md` §C3 asked for one. `TC-FRM-03`, `TC-FRM-04`, `TC-FRM-08` and `TC-NFR-06` reflect that absence honestly.
7. **A7: Defect identifiers.** Defects DEF-01 to DEF-13 are re-identified here from §B and §F4 of `docs/03-test-record.md`; that record uses B1–B11 and prose headings rather than ID prefixes. The cross-reference is one-to-one and stated in the table.